

L-convex polyominoes and a colored-partition comparison

A proof of the A126764 asymptotic, three correction terms,
an all-orders radial expansion, and inverse asymptotics

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Abstract

Let a_n count fixed L-convex polyominoes of area n , with $a_0 = 1$. Starting from the established area generating function, we prove the Guttmann–Kotěšovec asymptotic recorded as a conjecture in OEIS A126764:

$$a_n \sim \frac{13\sqrt{2}}{768 n^{3/2}} \exp\left(\pi\sqrt{\frac{13n}{6}}\right).$$

The main device is a positive comparison with a colored-partition Euler product $R(q)$, given explicitly below. If $p_n = [q^n]R(q)$, we obtain the finite, coefficientwise bound $0 \leq p_n - 4a_n \leq 3(p_n - 2p_{n-1} + p_{n-2})$ for every $n \geq 3$. A higher finite-cut identity then proves three explicit correction terms, with relative remainder $O(n^{-7/4})$. Independently, a finite recurrence computes the complete radial Poincaré expansion of $\mathcal{A}(e^{-t})$ to any prescribed order. We also derive an asymptotic inverse, including the necessary distinction between a real interpolation and an integer threshold. Exact integer checks through $n = 2000$ and symbolic coefficient checks accompany the manuscript. No numerical fit enters the proofs.

Scope and status. The proofs below are conventional mathematical proofs, not Lean certificates or a claim of independent peer review. The exact enumerative generating function is a cited input; its asymptotic analysis is supplied here. The all-orders *radial* expansion is proved. An all-orders *coefficient* expansion, and exponentially small transseries sectors for a_n , are not claimed.

Contents

1	The problem and the main results	3
1.1	The sequence and the source boundary	3
1.2	Constants and coefficient asymptotics	3
1.3	What the all-orders statement means	4
2	An exact recurrence with a positive normalization	5
3	Limiting slopes and the exact Euler product	6
3.1	Existence and analyticity of the slopes	6
3.2	Solving the auxiliary generating function	7
4	A coefficientwise proof of the leading conjecture	7
4.1	An exact positive defect	7
4.2	A positive majorant for the defect	7
4.3	Why the squeeze is asymptotically sharp enough	8
5	Finite cuts: an exact rational approximation	9
5.1	General positive remainder series	9
5.2	Eliminating the auxiliary evaluations	10
6	Saddle estimates for the comparison product	10
6.1	The local modular expansion	10
6.2	A minor-arc bound rather than a Tauberian shortcut	11
6.3	The transfer calculation	11
7	Proof of the three correction terms	12
7.1	Bounding the finite-cut remainder on the whole circle	12
7.2	Extracting the coefficients	13
8	A proved all-orders radial expansion	13
8.1	A compatible family of rational approximants	13
8.2	Why this is not yet an all-orders coefficient theorem	14
9	Inverse asymptotics and integer thresholds	14
9.1	A real inverse requires an interpolation convention	14
9.2	Lambert W and the correction terms	15
10	Reproducible exact and numerical checks	16
10.1	What was computed	16
10.2	Accuracy of the asymptotic approximants	16
10.3	Testing the inverse at known sequence values	17
11	Further research questions and concrete next steps	17
12	Conclusion and claim ledger	18
A	Reproduction commands and files	19
B	A short review checklist	19

1 The problem and the main results

1.1 The sequence and the source boundary

An L-convex polyomino is a convex polyomino in which any two cells can be joined inside the polyomino by a path with at most one turn. We count fixed polyominoes: translations are identified, but rotations and reflections are not generally identified. This is the convention in OEIS A126764 [1]. The initial terms are

$$1, 1, 2, 6, 15, 35, 76, 156, 310, 590, 1098, 1984, 3515, 6094, \dots$$

The zero-area term is a generating-function convention.

Castiglione, Frosini, Munarini, Restivo, and Rinaldi established an area generating function [2]; we use its explicit recurrence form as reproduced in Guttmann and Kotěšovec [3], equation (1.1). The latter authors proposed the leading asymptotic from a numerical analysis. The OEIS entry inspected for this report still labels that formula a conjecture [1]. A targeted search did not locate an earlier proof of the specific area asymptotic. This search is not a comprehensive priority certification.

The ProveIt repository [7] was inspected as the proposed research context. A targeted repository search for A126764 returned no matching file. The present manuscript is a standalone contribution: it does not assume any unverified theorem from another repository report. Enumeration by perimeter, and the asymptotics of 201-avoiding ascent sequences discussed alongside L-convex polyominoes in [3], are separate questions and are not asserted to be solved here.

[write] Added 2 October 2026, batch 75. This report is a single manuscript, printed in full: manuscript 05 of batch 75 of ProveIt's incoming-reports intake, delivered as the archive A126764_Lconvex_Research.zip in commit 4b874cea0 (main file lconvex_asymptotics.tex, now article.tex) and placed in commit 6ea60e367 in the research-report collection, among the OEIS-sequence asymptotics reports. The manuscript names no ProveIt commit: its repository input was an overview and a targeted search for A126764 on 1 October 2026, so it has no pin. That search is still accurate for the repository it saw; the only match now is this report. Its PDF metadata records it as "prepared with ChatGPT". Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. The writing step changed only the labels (every label now carries the prefix lcp:), set the bibliography ragged-right, and added these [write] notes and two references [8, 9]; no statement, proof or number of the manuscript was altered. The polyomino report nearest in subject, polyomino-growth-finite-prefix-corrections [9], bounds Klarner's growth constant $\lambda \leq 4.498$ for *all* polyominoes (exponential growth, with a Lean development Combinatorics/Polyominoes/KlarnerConstant for the weaker bound 9047/2000); the L-convex class grows only like $e^{B\sqrt{n}}$, so the two constants are unrelated and no theorem is shared. The report skew-partition-continued-fractions touches parallelogram polyominoes (A006958) only.

1.2 Constants and coefficient asymptotics

Throughout, put

$$K = \frac{13\pi^2}{24}, \quad B = 2\sqrt{K} = \pi\sqrt{\frac{13}{6}}, \quad c = \frac{13\sqrt{2}}{768}. \quad (1.1)$$

The letter K denotes a constant, not a generating function.

Theorem 1.1 (Coefficient expansion). *As $n \rightarrow \infty$ through the positive integers,*

$$a_n = c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + \frac{e}{n} + \frac{f}{n^{3/2}} + O(n^{-7/4}) \right), \quad (1.2)$$

where

$$d = \frac{\sqrt{K}}{6} + \frac{3}{2\sqrt{K}} = \frac{B}{12} + \frac{3}{B}, \quad (1.3)$$

$$e = \frac{1}{2} + \frac{3}{4K} - \frac{35K}{72}, \quad (1.4)$$

$$f = \frac{755K^{3/2}}{1296} + \frac{175\sqrt{K}}{72} - \frac{5}{8\sqrt{K}}. \quad (1.5)$$

Numerically,

$$d = 1.0341052176262747\dots, \quad e = -1.9584764928967362\dots, \quad f = 12.550445779146218\dots$$

In particular, the leading asymptotic stated in the abstract holds.

There are two complementary proof routes. The first proves the leading formula and its $n^{-1/2}$ correction by an explicit positive squeeze. The second uses a finite rational approximation and a contour estimate; it proves all the terms in Theorem 1.1.

Theorem 1.2 (A finite coefficient squeeze). *Define*

$$R(q) = \frac{(-q; q^2)_\infty}{(q; q)_\infty^3} = \frac{(q^2; q^2)_\infty^2}{(q; q)_\infty^4 (q^4; q^4)_\infty} = \sum_{n \geq 0} p_n q^n, \quad J(q) = (1 - q)^2 R(q) = \sum_{n \geq 0} j_n q^n. \quad (1.6)$$

Then, for every $n \geq 3$,

$$0 \leq p_n - 4a_n \leq 3j_n = 3(p_n - 2p_{n-1} + p_{n-2}). \quad (1.7)$$

Both R and J have nonnegative integer coefficients. Moreover,

$$n \left(1 - \frac{4a_n}{p_n} \right) \rightarrow \frac{K}{2} = \frac{13\pi^2}{48}. \quad (1.8)$$

The product R counts partitions with four colors available for an odd part, two for a part congruent to 2 modulo 4, and three for a multiple of 4. Thus the constant 13 comes from the average number $13/4$ of colors, not from a fit to a decimal growth constant. The factor $1/4$ in the comparison is forced by two boundary conditions of the recurrence. We do not assert a four-to-one combinatorial map; finding one is a natural further problem.

1.3 What the all-orders statement means

Let $t > 0$ and $q = e^{-t}$. We will prove

$$\mathcal{A}(e^{-t}) \sim \frac{t^{3/2}}{4(2\pi)^{3/2}} \exp\left(\frac{K}{t} - \frac{t}{6}\right) \mathcal{F}(t), \quad (1.9)$$

where $\mathcal{F}$ is a formally specified series beginning

$$\begin{aligned} \mathcal{F}(t) = & 1 - \frac{1}{2}t^2 + \frac{1}{2}t^3 - \frac{41}{48}t^4 + \frac{7}{4}t^5 - \frac{6317}{1440}t^6 \\ & + \frac{1529}{120}t^7 - \frac{1702513}{40320}t^8 + \dots \end{aligned} \quad (1.10)$$

Here $\sim$ is meant to all fixed algebraic orders along the positive real axis. It is not a claim that this series converges, nor that a real-axis expansion alone implies coefficient asymptotics. The complex estimate needed for the three coefficient corrections is proved separately.

2 An exact recurrence with a positive normalization

Write $(a; q)_m = \prod_{j=0}^{m-1} (1 - aq^j)$, and set $(q; q)_0 = 1$. The enumerative input is

$$\mathcal{A}(q) = 1 + \sum_{n \geq 1} \frac{q^n h_n(q)}{(q; q)_{n-1}^2 (1 - q^n)}, \quad (2.1)$$

where

$$h_0 = 1, \quad h_1 = 1, \quad h_{n+1} = 2h_n - (1 - q^n)^2 h_{n-1} \quad (n \geq 1). \quad (2.2)$$

Indeed, $h_n = f_{n-1}$ for $n \geq 1$ in the notation of [3], and $h_2 = 1 + 2q - q^2$. Introducing h_0 makes the recurrence valid already at $n = 1$.

The auxiliary solution

$$Q_0 = 1, \quad Q_1 = 2, \quad Q_{n+1} = 2Q_n - (1 - q^n)^2 Q_{n-1} \quad (2.3)$$

will produce the comparison product. For a general solution y_n of the same recurrence, define

$$b_n = \frac{y_n}{(q; q)_n^2}, \quad e_n = b_n - b_{n-1}, \quad T_n(q) = \frac{1}{(1 - q^n)^2} - 1.$$

We reserve d, e, f without subscripts for the constants in Theorem 1.1.

Lemma 2.1 (Positive transfer). *For $n \geq 2$,*

$$\begin{aligned} b_n &= \frac{b_{n-1} + e_{n-1}}{(1 - q^n)^2}, \\ e_n &= e_{n-1} + T_n(q)(b_{n-1} + e_{n-1}). \end{aligned} \quad (2.4)$$

Every entry of this linear transfer rule is a power series with nonnegative coefficients. Consequently, coefficientwise inequalities between two pairs (b_m, e_m) propagate to all later pairs and to their limiting slopes, whenever the slopes exist.

Proof. Dividing the recurrence by $(q; q)_{n-1}^2$ gives $(1 - q^n)^2 b_n = 2b_{n-1} - b_{n-2} = b_{n-1} + e_{n-1}$. Subtract b_{n-1} . Finally, $T_n = \sum_{r \geq 1} (r+1)q^{nr}$ has nonnegative coefficients. $\square$

For h , the initial normalized pair is

$$b_1 = (1 - q)^{-2}, \quad e_1 = (1 - q)^{-2} - 1,$$

and both entries are nonnegative. The same holds for Q , with $b_1 = 2(1 - q)^{-2}$ and $e_1 = 2(1 - q)^{-2} - 1$. Throughout, $F \preccurlyeq G$ means that every coefficient of $G - F$ is nonnegative. This is a stronger relation than $F(q) \leq G(q)$ for $0 < q < 1$. We will be careful not to multiply a coefficientwise inequality by a polynomial with negative coefficients.

3 Limiting slopes and the exact Euler product

3.1 Existence and analyticity of the slopes

Lemma 3.1. *For a solution with polynomial initial values and every $0 < r < 1$, there are analytic functions $s(q), b(q)$ on $|q| < 1$ such that, uniformly on $|q| \leq r$,*

$$y_n(q) = s(q)n + b(q) + O_r(n^2 r^n).$$

In particular, the normalized limiting slope

$$\mathcal{L}_y(q) = \lim_{n \rightarrow \infty} \left(\frac{y_n}{(q; q)_n^2} - \frac{y_{n-1}}{(q; q)_{n-1}^2} \right) = \frac{s(q)}{(q; q)_\infty^2} \quad (3.1)$$

exists locally uniformly. The same limit exists coefficientwise as a formal power series.

Proof. Put $v_j = 2q^j - q^{2j}$. Twice summing $y_{j+1} - 2y_j + y_{j-1} = v_j y_{j-1}$ gives

$$y_n = y_0 + n(y_1 - y_0) + \sum_{j=1}^{n-1} (n-j)v_j y_{j-1}.$$

On $|q| \leq r$, $|v_j| \leq 3r^j$. Divide by $n+1$ and apply the elementary finite-product version of discrete Gronwall's inequality. The resulting bound is a constant times $\prod_{j \geq 1} (1 + 3jr^j) < \infty$. Thus $y_n = O_r(n+1)$. The increments $y_{n+1} - y_n$ have a uniformly convergent sum of successive differences, with exponentially small tails. One more summation yields the displayed affine expansion. Since $(q; q)_n / (q; q)_\infty = 1 + O_r(r^n)$, normalization gives (3.1). Finally, in (2.4) the change in e_n has valuation at least n . Each fixed coefficient therefore stabilizes. $\square$

Write $L = \mathcal{L}_h$. If

$$C(z; q) = \sum_{n \geq 0} \frac{h_n(q)}{(q; q)_n^2} z^n, \quad C_j(q) = C(q^j; q), \quad (3.2)$$

then C has radius of convergence at least 1 in z and a possible double pole at $z = 1$ with leading coefficient L . Equation (2.1) becomes

$$\mathcal{A} = 1 + C_1 - C_2. \quad (3.3)$$

Multiplying the normalized recurrence by z^n and summing, including the initial condition $h_1 = 1$, yields

$$(1-z)^2 C(z; q) - 2C(qz; q) + C(q^2 z; q) = -z. \quad (3.4)$$

Taking $z \rightarrow 1$ is now legitimate and gives

$$L = 2C_1 - C_2 - 1, \quad \mathcal{A} = L - C_1 + 2. \quad (3.5)$$

The pole term must not be dropped in this limit.

3.2 Solving the auxiliary generating function

Define $G(z) = \sum_{n \geq 0} Q_n z^n / (q; q)_n$. Directly from (2.3),

$$(1 - z)^2 G(z) = (1 + qz^2) G(qz), \quad G(0) = 1.$$

Iteration gives the exact identity

$$G(z) = \frac{(-qz^2; q^2)_\infty}{(z; q)_\infty^2}. \quad (3.6)$$

At $z = 1$, the coefficient of $(1 - z)^{-2}$ is $(-q; q^2)_\infty / (q; q)_\infty^2$. It follows, either by subtracting the principal part or by Lemma 3.1, that

$$\mathcal{L}_Q = \frac{(-q; q^2)_\infty}{(q; q)_\infty^3} = R(q). \quad (3.7)$$

The second expression for R in (1.6) follows from $(-q; q^2)_\infty = (q^2; q^2)_\infty^2 / ((q; q)_\infty (q^4; q^4)_\infty)$. In particular,

$$R(q) = \prod_{j \geq 1} (1 - q^j)^{-b_j}, \quad b_j = 4 - 2\mathbf{1}_{2|j} + \mathbf{1}_{4|j}. \quad (3.8)$$

This proves the colored-partition interpretation stated earlier.

4 A coefficientwise proof of the leading conjecture

4.1 An exact positive defect

Let $v_n = Q_n - 2h_n$. Then $v_0 = -1$, $v_1 = 0$, $v_2 = (1 - q)^2$. Its normalized pair at index 2 is

$$b_2 = e_2 = (1 - q^2)^{-2}.$$

The positive transfer lemma shows that its slope

$$E = R - 2L \quad (4.1)$$

has nonnegative coefficients. Combining (3.3) and (3.5) gives the exact decomposition

$$R - 4\mathcal{A} + 6 = E + 2C_2. \quad (4.2)$$

Both terms on the right are nonnegative power series. The constant 6 only corrects the chosen zero-area terms.

4.2 A positive majorant for the defect

We prove the bounds

$$0 \preccurlyeq E \preccurlyeq J, \quad 0 \preccurlyeq C_2 \preccurlyeq J + q + 2q^2. \quad (4.3)$$

These statements require a little care: a real-axis bound by $(1 - q)^2 R$ would not by itself imply a bound on coefficients.

First multiply the normalized Q solution by $(1 - q)^2$, and denote the resulting pair by $(\tilde{b}_n, \tilde{e}_n)$. At $n = 2$,

$$\begin{aligned}\tilde{b}_2 &= \frac{3 + 2q - q^2}{(1 - q^2)^2}, \\ \tilde{e}_2 &= \frac{1 + 2q + 3q^2 - 2q^4}{(1 - q^2)^2}.\end{aligned}$$

Subtracting the normalized pair for v gives

$$\frac{2 + 2q - q^2}{(1 - q^2)^2}, \quad \frac{2q + 3q^2 - 2q^4}{(1 - q^2)^2}.$$

Both are nonnegative power series: use $2 + 2q - q^2 = (1 + 2q) + (1 - q^2)$ and $3q^2 - 2q^4 = q^2 + 2q^2(1 - q^2)$. The transfer lemma and passage to slopes prove $E \preccurlyeq J$. The displayed pair for $\tilde{b}_2, \tilde{e}_2$ itself is nonnegative by the same decompositions.

Next, every coefficient of $(1 - q^n)h_n/(q; q)_n^2$ is nonnegative. For $n = 1$ this is immediate; for $n \geq 2$ it follows by removing only one of the two factors $(1 - q^n)$ in the denominator in (2.4). Hence $C_2 - C_3 \geq 0$. Setting $z = q$ in (3.4) gives

$$2C_2 - C_3 = (1 - q)^2 C_1 + q, \quad C_2 \preccurlyeq (1 - q)^2 C_1 + q.$$

It remains to compare the last expression with J .

Let $C_j^Q = \sum_{n \geq 0} Q_n q^{jn} / (q; q)_n^2$. For the Q solution the initial-condition term in (3.4) is zero, so $R = 2C_1^Q - C_2^Q$. The solution $u_n = Q_n - h_n$ has $u_0 = 0, u_1 = 1$; after normalization and multiplication by $(1 - q)^2$ its initial pair at 1 is $(1, 1)$. Consequently,

$$(1 - q)^2 (C_1^Q - C_1) \geq 0.$$

Also

$$(1 - q)^2 (C_1^Q - C_2^Q) = \sum_{n \geq 1} q^n (1 - q^n) \tilde{b}_n \geq -2q^2.$$

For completeness, the $n = 1$ summand is $2q - 2q^2$. For $n = 2$ the remaining factor is $(3 + 2q - q^2)/(1 - q^2) = 1 + (2 + 2q)/(1 - q^2)$, which is nonnegative. For $n \geq 3$, positivity follows from the nonnegative pair at index 2 and (2.4). These two inequalities imply $(1 - q)^2 C_1 \preccurlyeq J + 2q^2$, establishing (4.3).

Taking coefficients of (4.2) now proves (1.7). In particular, the finite bound is independent of any asymptotic approximation.

4.3 Why the squeeze is asymptotically sharp enough

The analytic estimates in Section 6 imply

$$p_n = 4c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + O(n^{-1}) \right), \quad (4.4)$$

$$j_n \sim \frac{K}{n} p_n. \quad (4.5)$$

Therefore the squeeze alone gives

$$a_n = c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + O(n^{-1}) \right).$$

This already proves the full proposed leading constant and the first correction. The further work below is needed for e and f , not for the leading conjecture.

5 Finite cuts: an exact rational approximation

5.1 General positive remainder series

The discrete Wronskian of h and Q is

$$h_m Q_{m+1} - h_{m+1} Q_m = (q; q)_m^2. \quad (5.1)$$

At $m = 0$ both sides are 1; the recurrence multiplies each succeeding Wronskian by $(1 - q^m)^2$, proving the identity by induction. Define

$$E_m = h_m R - Q_m L \quad (m \geq 1). \quad (5.2)$$

Thus $E_1 = E$.

Lemma 5.1 (A positive tail with a small radial bound). *The series E_m has nonnegative coefficients. For real $0 < q < 1$,*

$$0 \leq E_m(q) \leq (q; q)_m^2 R(q). \quad (5.3)$$

Furthermore, for each fixed m ,

$$C_{m+1}(e^{-t}) = O_m(t^{2m} R(e^{-t}) + 1) \quad (t \downarrow 0). \quad (5.4)$$

Proof. For $n \geq m$, the solution $w_n = h_m Q_n - Q_m h_n$ has $w_m = 0$, $w_{m+1} = (q; q)_m^2$. Its normalized pair at $m+1$ is $(1 - q^{m+1})^{-2}$ in both coordinates. Hence it is positive and its limiting slope, E_m , is positive.

For the real bound, let $z_m = 0$, $z_{m+1} = 1$ satisfy the original recurrence for later indices. Then $w_n = (q; q)_m^2 z_n$. For real q , the increments of Q_n are positive and nondecreasing:

$$Q_{n+1} - Q_n = (Q_n - Q_{n-1}) + (2q^n - q^{2n})Q_{n-1}.$$

Thus $Q_n \geq n + 1$ and $\Delta Q_m = Q_m - Q_{m-1} \geq 1$. A second Wronskian calculation yields

$$\frac{s_z}{s_Q} = Q_m \sum_{k=m}^{\infty} \frac{\prod_{j=m+1}^k (1 - q^j)^2}{Q_k Q_{k+1}} \leq \frac{1}{\Delta Q_m} \leq 1.$$

To obtain the middle inequality, use $Q_k \geq Q_m + (k - m)\Delta Q_m$ and telescope the reciprocals of the resulting arithmetic progression. Since normalization divides both slopes by $(q; q)_{\infty}^2$, (5.3) follows.

Finally, positivity gives $C_1 \leq C_1^Q \leq R$ on the real axis. Indeed, $Q - h$ is positive after normalization, and $R - C_1^Q = C_1^Q - C_2^Q \geq 0$. From (3.4) and $C_{j+1} \geq C_{j+2}$ on the real axis,

$$C_{j+1} \leq (1 - q^j)^2 C_j + q^j.$$

Iterate this inequality a fixed number m of times. The leading coefficient is $(q; q)_m^2 = O_m(t^{2m})$, and the finitely many remaining terms are bounded as $t \downarrow 0$. $\square$

5.2 Eliminating the auxiliary evaluations

Put $U_m = Q_m - h_m$, and let polynomials P_m be specified by

$$P_0 = P_1 = 0, \quad P_{j+1} = 2P_j - (1 - q^j)^2 P_{j-1} + q^j.$$

Iteration of (3.4) gives

$$C_{m+1} = Q_m C_1 - U_m(L + 1) - P_m.$$

Solving this identity together with (5.2) and $\mathcal{A} = L - C_1 + 2$ proves the following exact formula.

Proposition 5.2 (Finite-cut identity). *Where $Q_m(q) \neq 0$,*

$$\mathcal{A}(q) = \left(\frac{h_m}{Q_m} \right)^2 R - \frac{h_m E_m}{Q_m^2} - \frac{C_{m+1}}{Q_m} + 2 - \frac{U_m + P_m}{Q_m}. \quad (5.5)$$

After multiplication by Q_m^2 , this is an identity of analytic functions on the entire open unit disk and of formal power series.

The square in the first term is the precise source of the factor $1/4$: for every fixed $m \geq 1$, $h_m(1) = 2^{m-1}$ and $Q_m(1) = 2^m$. The identity also provides an algebraic expansion algorithm and a route to rigorous coefficient bounds.

For the coefficient theorem we use only $m = 3$:

$$\begin{aligned} h_3 &= 1 + 4q - q^4, & Q_3 &= 4 + 4q + 2q^2 - 2q^4 = -2(q+1)(q^3 - q^2 - 2), \\ U_3 + P_3 &= 3 + 2q + 3q^2 - q^4, & F_3(q) &= \left(\frac{1 + 4q - q^4}{4 + 4q + 2q^2 - 2q^4} \right)^2. \end{aligned} \quad (5.6)$$

The cubic $q^3 - q^2 - 2$ has no zero in $|q| \leq 1$. A zero there would satisfy $|q|^2|q - 1| = 2$; equality in the bound $|q|^2|q - 1| \leq 2$ forces $q = -1$, which does not solve the equation. Thus $F_3 R$ is analytic in $|q| < 1$, with only a possible additional boundary pole at $q = -1$.

6 Saddle estimates for the comparison product

6.1 The local modular expansion

The classical eta transformation, in its Euler-product form, is

$$(e^{-z}; e^{-z})_\infty = \sqrt{\frac{2\pi}{z}} \exp\left(-\frac{\pi^2}{6z} + \frac{z}{24}\right) (e^{-4\pi^2/z}; e^{-4\pi^2/z})_\infty, \quad \Re z > 0. \quad (6.1)$$

The square root is the branch positive for positive real z . This follows from the usual modular transformation of Dedekind's eta function; see [5], equations 27.14.12–27.14.14. Applying it at $z, 2z, 4z$ gives, uniformly in any fixed closed sector $|\arg z| < \pi/2$,

$$R(e^{-z}) = \left(\frac{z}{2\pi} \right)^{3/2} \exp\left(\frac{K}{z} - \frac{z}{6}\right) \left(1 + O(e^{-\kappa/|z|})\right) \quad (6.2)$$

for an appropriate sector-dependent $\kappa > 0$. In particular, there is no omitted algebraic power-series correction in (6.2) beyond $\exp(-z/6)$.

6.2 A minor-arc bound rather than a Tauberian shortcut

Let $r = e^{-t}$. The colored-product form (3.8) implies

$$\frac{|R(re^{i\theta})|}{R(r)} \leq \exp\left[-\frac{c_0}{t} \min\left\{1, \frac{\theta^2}{t^2}\right\}\right] \quad (-\pi \leq \theta \leq \pi) \quad (6.3)$$

for all sufficiently small t and some $c_0 > 0$. Here and below a constant such as c_0 is unrelated to c in (1.1).

To verify (6.3), restrict the product to integers $N \leq j \leq 2N$, with N comparable to $1/t$. Then r^j stays in a compact subinterval of $(0, 1)$, and

$$\log \frac{|1 - r^j e^{ij\theta}|}{1 - r^j} \geq c_1(1 - \cos(j\theta)).$$

In the product for $|R(re^{i\theta})|/R(r)$, every omitted factor is at most 1. The elementary trigonometric-sum estimate

$$\sum_{j=N}^{2N} (1 - \cos(j\theta)) \geq c_2 N \min\{1, N^2 \theta^2\}$$

finishes the proof. For $|\theta| \leq 1/N$ it follows from the quadratic bound for $1 - \cos x$; for an intermediate compact range of $N\theta$ it follows by comparison with its strictly positive integral; for larger $N\theta$ use the geometric-sum bound for $\sum e^{ij\theta}$. This argument also covers the neighborhood of $\theta = \pi$.

Consequently a rational factor with at most fixed-order poles on the unit circle does not create a competing dominant saddle. The product's exponential suppression outweighs such polynomial losses.

6.3 The transfer calculation

For a local expansion

$$H(e^{-z}) \sim \lambda z^{3/2} e^{K/z} \sum_{j \geq 0} s_j z^j,$$

with the preceding minor-arc control, Cauchy's integral is localized at $t = \sqrt{K/n}$ and angular width $t^{3/2}$. Termwise saddle integration gives

$$[q^n]H(q) \sim \frac{\lambda K}{2\sqrt{\pi}} n^{-3/2} e^{2\sqrt{Kn}} \sum_{\ell \geq 0} n^{-\ell/2} \sum_{j+k=\ell} s_j K^{j/2} \frac{\beta_k(j)}{(2\sqrt{K})^k}, \quad (6.4)$$

where

$$\beta_k(j) = \frac{(-1)^k}{k! 8^k} \prod_{r=1}^k ((5+2j)^2 - (2r-1)^2), \quad \beta_0(j) = 1. \quad (6.5)$$

For clarity, (6.4) asserts a finite-order expansion whenever a corresponding finite-order local expansion is available.

Here is a direct way to check the constants. The model integral for $z^{3/2+j} e^{K/z}$ is the Bessel-type contour expression

$$\left(\frac{K}{n}\right)^{(5/2+j)/2} I_{-5/2-j}(2\sqrt{Kn}).$$

Its large positive-argument saddle expansion is $e^x/\sqrt{2\pi x}$ times the polynomial coefficients (6.5) in powers of x^{-1} . Alternatively those same coefficients follow by substituting a power series into the Bessel differential equation. The coefficient $\lambda K/(2\sqrt{\pi})$ in (6.4) follows immediately. One need not identify the original generating function with a Bessel function: only this local contour model is used.

For a fully controlled remainder, take a central arc $|\theta| \leq t^{3/2-\delta}$ with $0 < \delta < 1/6$. On the complement, (6.3) makes the integral smaller than any algebraic multiple of its leading saddle value. On the central arc, put $\theta = t^{3/2}u$; the real part of the quadratic term is a negative constant times u^2 , and Taylor remainders are bounded by a polynomial in u times that Gaussian. Truncation and integration are therefore valid to every fixed order. This supplies the error justification behind (6.4).

For $H = R$, use $\lambda = (2\pi)^{-3/2}$ and $s_j = (-1/6)^j/j!$. This proves (4.4). Multiplication by $(1 - e^{-z})^2 = z^2 + O(z^3)$ gives (4.5). It also gives

$$p_n = 4c n^{-3/2} e^{B\sqrt{n}} \left(1 - \frac{d}{\sqrt{n}} + \frac{\frac{1}{2} + \frac{3}{4K} + \frac{K}{72}}{n} + O(n^{-3/2}) \right). \quad (6.6)$$

The comparison product itself has an all-orders coefficient expansion; this fact is not being inferred for $\mathcal{A}$ without additional estimates.

7 Proof of the three correction terms

7.1 Bounding the finite-cut remainder on the whole circle

Proposition 7.1. *With F_3 from (5.6),*

$$a_n = [q^n](F_3 R) + O(p_n n^{-7/4}). \quad (7.1)$$

Proof. Take $q = re^{i\theta}$ with $r = e^{-t}$ and $t = \sqrt{K/n}$. By positivity and Lemma 5.1,

$$|E_3(q)| \leq E_3(r) = O(t^6 R(r)), \quad |C_4(q)| \leq C_4(r) = O(t^6 R(r) + 1).$$

Since the cubic factor of Q_3 is bounded away from zero on the closed unit disk, there are uniform estimates

$$\begin{aligned} \int_{-\pi}^{\pi} \frac{d\theta}{|Q_3(re^{i\theta})|^2} &= O(t^{-1}), \\ \int_{-\pi}^{\pi} \frac{d\theta}{|Q_3(re^{i\theta})|} &= O(\log(1/t)). \end{aligned}$$

The first follows from $\int_{-\pi}^{\pi} |1 + re^{i\theta}|^{-2} d\theta = 2\pi/(1 - r^2)$; the second follows by integrating $(t^2 + (\theta - \pi)^2)^{-1/2}$ near the single boundary zero. The polynomials h_3, U_3, P_3 are uniformly bounded.

Apply Cauchy's coefficient formula to the three remainder terms in (5.5) with $m = 3$. Their total is

$$O(r^{-n} \{t^5 R(r) + t^6 R(r) \log(1/t) + \log(1/t) + 1\}).$$

The terms without $R(r)$ are exponentially smaller than the main coefficient scale. Since the saddle estimate gives $p_n \asymp r^{-n} R(r) t^{3/2}$, the dominant error is $O(p_n t^{7/2}) = O(p_n n^{-7/4})$. $\square$

7.2 Extracting the coefficients

A finite rational expansion at $q = 1$ gives

$$4F_3(e^{-z}) = 1 - \frac{1}{2}z^2 + \frac{1}{2}z^3 + O(z^4).$$

Together with (6.2), this yields

$$(F_3R)(e^{-z}) = \frac{z^{3/2}e^{K/z}}{4(2\pi)^{3/2}} \left(1 - \frac{z}{6} - \frac{35z^2}{72} + \frac{755z^3}{1296} + O(z^4) \right), \quad (7.2)$$

up to a locally exponentially small modular remainder. The rational factor has only fixed-order boundary poles, so the minor-arc argument applies. Using (6.4) through $\ell = 3$ gives (1.3)–(1.5); its own truncation error is $O(n^{-2})$ relative to the leading term. The larger remainder in Proposition 7.1 is $O(n^{-7/4})$, proving Theorem 1.1.

For an explicit audit of the algebra, the relative n^{-1} coefficient is

$$\underbrace{\frac{3}{4K}}_{j=0,k=2} + \underbrace{\frac{1}{2}}_{j=1,k=1} - \underbrace{\frac{35K}{72}}_{j=2,k=0}.$$

The relative $n^{-3/2}$ coefficient is

$$\underbrace{-\frac{5}{8\sqrt{K}}}_{j=1,k=2} + \underbrace{\frac{175\sqrt{K}}{72}}_{j=2,k=1} + \underbrace{\frac{755K^{3/2}}{1296}}_{j=3,k=0}.$$

There is no $j = 0, k = 3$ contribution: the Bessel polynomial for that half-integer order terminates after $k = 2$.

Subtracting four times (1.2) from (6.6) proves (1.8). Thus the lost proportion in the colored-partition comparison is asymptotically $K/(2n)$, substantially smaller than the explicit upper bound $3K/n$ supplied by the elementary squeeze.

8 A proved all-orders radial expansion

8.1 A compatible family of rational approximants

Define $D(q) = L(q)/R(q)$ for real $0 < q < 1$. For every fixed $m \geq 1$, (5.2) and (5.3) imply

$$D(e^{-t}) = \frac{h_m(e^{-t})}{Q_m(e^{-t})} + O_m(t^{2m}). \quad (8.1)$$

The denominator tends to 2^m , so it is nonzero near the positive real limit. The Taylor expansions of these rational functions are therefore compatible: the coefficient of t^j is fixed once $2m > j$. Let $\mathcal{D}(t)$ be their unique formal limit.

Theorem 8.1 (Radial expansion to arbitrary fixed order). *The expansion (1.9) holds with*

$$\mathcal{F}(t) = 4\mathcal{D}(t)^2, \quad \mathcal{D}(t) = \lim_{m \rightarrow \infty} \frac{\overset{\text{formal}}{h_m}(e^{-t})}{Q_m(e^{-t})}. \quad (8.2)$$

For each fixed m ,

$$\frac{\mathcal{A}(e^{-t})}{R(e^{-t})} = \left(\frac{h_m(e^{-t})}{Q_m(e^{-t})} \right)^2 + O_m(t^{2m}). \quad (8.3)$$

The coefficients through t^8 are those in (1.10).

Proof. Divide the finite-cut identity (5.5) by R . For fixed m , the denominator $Q_m(e^{-t})$ stays bounded away from zero. The E_m/R and C_{m+1}/R terms are $O_m(t^{2m})$ by Lemma 5.1; the remaining rational expression divided by R is smaller than every power of t by (6.2). This proves (8.3). Compatibility and the product expansion give (8.2). Ordinary rational series arithmetic then gives the displayed coefficients. $\square$

For example, setting $w = 1 - q$ gives the useful intermediate expression

$$\mathcal{D} = \frac{1}{2} - \frac{1}{8}w^2 - \frac{5}{32}w^4 + \frac{1}{8}w^5 - \frac{65}{128}w^6 + \frac{37}{32}w^7 - \frac{2053}{512}w^8 + O(w^9).$$

Replacing w by $1 - e^{-t}$ and squaring reproduces (1.10). The accompanying symbolic checker verifies these operations exactly over the rationals.

8.2 Why this is not yet an all-orders coefficient theorem

Increasing m makes the positive radial tails smaller, but the complex behavior of $1/Q_m$ must still be controlled. For general m , one cannot simply assume that all zeros of Q_m lie outside the unit disk. No general zero classification is assumed or established here. A meromorphic rational approximant can be useful on a contour, but poles inside that contour require attention if one interprets its integral as an ordinary Taylor coefficient.

The choice $m = 3$ deliberately avoids these issues: the factorization in (5.6) proves the required zero exclusion exactly. Consequently the first three correction terms are supported by a full complex argument. Coefficients at later orders obtained by purely formal transfer of (1.9) are candidates until an equally valid remainder argument is supplied. No exponentially small contribution from other roots of unity, nor a complete transseries with all such sectors, is claimed in this report.

9 Inverse asymptotics and integer thresholds

9.1 A real inverse requires an interpolation convention

The sequence itself is defined only on integers. To state a real inverse accurately, choose an interpolation that preserves the proved asymptotic accuracy. Put

$$M(x) = c x^{-3/2} e^{B\sqrt{x}} (1 - dx^{-1/2} + ex^{-1} + fx^{-3/2})$$

for sufficiently large real x , where the parenthesis is positive. For an integer k , set $r_k = \log a_k - \log M(k)$. Theorem 1.1 gives $r_k = O(k^{-7/4})$. For $x = k + \theta$, $0 \leq \theta \leq 1$, define

$$\log \tilde{a}(x) = \log M(x) + (1 - \theta)r_k + \theta r_{k+1}. \quad (9.1)$$

This continuous interpolation agrees with a_k at integers and obeys

$$\log \tilde{a}(x) = \log M(x) + O(x^{-7/4}).$$

It is eventually strictly increasing: the derivative of $\log M$ is $B/(2\sqrt{x}) + O(x^{-1})$, while the piecewise-linear residual has slope $O(x^{-7/4})$. Denote its eventual inverse by $v(y)$. Other interpolations with the same asymptotic error give the same expansion below to the stated accuracy.

This convention matters beyond low orders. Direct linear interpolation of $\log a_k$, without first subtracting $\log M$, introduces a curvature error of order $x^{-3/2}$, which can obscure the third logarithmic correction. Equation (9.1) avoids that ambiguity.

9.2 Lambert W and the correction terms

Define

$$\begin{aligned} D_0 &= Bd = \frac{B^2}{12} + 3, \\ \alpha_2 &= B^2(e - d^2/2), \quad \alpha_3 = B^3(f + de - d^3/3), \\ \ell &= \log \frac{y}{cB^3}, \quad s_0 = -3W_{-1}\left(-\frac{1}{3}e^{-\ell/3}\right). \end{aligned} \tag{9.2}$$

Here W_{-1} is the real lower branch, which tends to $-\infty$ as its argument tends to 0 from below. It is selected because the inverse index tends to $+\infty$.

Theorem 9.1 (Asymptotic inverse). *For the interpolation (9.1), as $y \rightarrow \infty$,*

$$v(y) = \frac{1}{B^2} \left[s_0^2 + 2D_0 + \frac{2(3D_0 - \alpha_2)}{s_0} + \frac{18D_0 - 6\alpha_2 - D_0^2 - 2\alpha_3}{s_0^2} \right] + O(s_0^{-5/2}). \tag{9.3}$$

In particular, a formula involving only logarithms is

$$v(y) = \frac{\ell^2 + 6\ell \log \ell + 9(\log \ell)^2 + 18 \log \ell + 2D_0}{B^2} + O\left(\frac{(\log \ell)^2}{\ell}\right). \tag{9.4}$$

Proof. Set $s = B\sqrt{v(y)}$. Taking logarithms in the coefficient expansion and using the interpolation error gives

$$\ell = s - 3 \log s - \frac{D_0}{s} + \frac{\alpha_2}{s^2} + \frac{\alpha_3}{s^3} + O(s^{-7/2}). \tag{9.5}$$

The definition of s_0 is equivalent to $s_0 - 3 \log s_0 = \ell$. Put $s = s_0 + \delta$ and compare successive inverse powers of s_0 . Since $1 - 3/s_0$ is bounded away from zero,

$$\delta = \frac{D_0}{s_0} + \frac{3D_0 - \alpha_2}{s_0^2} + \frac{9D_0 - 3\alpha_2 - D_0^2 - \alpha_3}{s_0^3} + O(s_0^{-7/2}).$$

Squaring and dividing by B^2 gives (9.3). For (9.4), direct substitution in (9.5) yields

$$s = \ell + 3 \log \ell + \frac{9 \log \ell + D_0}{\ell} + O((\log \ell)^2/\ell^2),$$

and another squaring completes the calculation. □

If $N(y) = \min\{n \geq n_0 : a_n \geq y\}$ for a sufficiently large fixed n_0 , then exactly

$$N(y) = \lceil v(y) \rceil.$$

The error term belongs *inside* that ceiling. Uniformly discarding it can produce an incorrect integer when the approximation is very close to a threshold. A safe coarser statement is that the right-hand side of (9.4), without its remainder, differs from $N(y)$ by $O(1)$. Away from integers by more than a proved error bound, rounding would be unambiguous; this article does not supply a numerical universal constant for that error bound.

[write] **Added 2 October 2026, batch 75: an instance of repository results.** The inversion of this section is an instance of the Lambert-centred inversion apparatus of the transseries-and-inversion volume [8], and no novelty is claimed for the method. The core $s_0 - 3 \log s_0 = \ell$ is `p0:thm:lambert-core` with $a = 1$, $b = -3$; its branch rule selects W_{-1} , as above. Equation (9.5) is the core plus a power series in $1/s$, and the correction δ is the all-orders reversion around the core, `p0:thm:lambert-centered`, truncated at the order the forward remainder allows. The shape is that of the partition-number chapter of the same volume, `p3:sec:lambert` with `p3:thm:core-correction`, where the factor n^{-1} in $p(n)$ gives $b = -2$; here $n^{-3/2}$ gives $b = -3$. The rule $N(y) = \lceil v(y) \rceil$ and the warning that follows it are part (1) and part (2) (the separation condition) of `p0:thm:staircase`, specialized as in `p3:thm:staircase`. The generic arithmetic of those two parts is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; those statements concern an arbitrary monotone interpolation and give no formal status to Theorem 9.1. What is specific to this report is the forward expansion that feeds the inversion and the explicit interpolation (9.1).

10 Reproducible exact and numerical checks

10.1 What was computed

The attached `verify.py` uses only Python's standard library and unbounded integers. It computes $a_0, \dots, a_{2000}$ from the normalized recurrence, and independently computes p_n from the colored Euler product. Division by $(1 - q^m)^2$ is performed by two exact cumulative-sum passes. Thus the computation avoids floating-point cancellation in the original polynomial recurrence.

The checker confirms the initial 37 OEIS terms and four separately read b-file anchors, at $n = 50, 100, 200, 500$ [4]. It then verifies the slope identity, the positive defect decomposition, both majorants in (4.3), and the squeeze for every integer $3 \leq n \leq 2000$. The third finite-cut identity, after clearing its polynomial denominator, and positivity of its auxiliary tails are checked as well. The output CSV contains all computed coefficients, not just a selected plot. This is an exact finite test, not a formal proof of the infinite statements.

The optional `verify_symbolic.py` uses SymPy for rational symbolic arithmetic and mpmath for high-precision numerical evaluations. It checks Wronskians, the factorization of Q_3 , the radial coefficients through t^8 , and the transfer coefficients d, e, f . All included assertions passed in the supplied run.

10.2 Accuracy of the asymptotic approximants

Let

$$A_0(n) = cn^{-3/2}e^{B\sqrt{n}}, \quad A_1 = A_0(1 - d/\sqrt{n}), \quad A_2 = A_1 + A_0e/n, \quad A_3 = A_2 + A_0f/n^{3/2}.$$

The table reports $a_n/A_j(n)$; a ratio approaching 1 is the desired behavior. Rounded display values are not used by the checker.

n	a_n/A_0	a_n/A_1	a_n/A_2	a_n/A_3
50	0.836729727	0.980057860	1.027184142	0.984290725
100	0.885790139	0.987955090	1.010017535	0.995767525
200	0.920485946	0.993103974	1.003708062	0.998875075
500	0.950776376	0.996878614	1.000989560	0.999807957
1000	0.965688969	0.998335826	1.000361244	0.999950135
2000	0.976025253	0.999128395	1.000130943	0.999987161

More terms need not improve a short truncation at a modest index: $n = 50$ illustrates this. At $n = 2000$, the four-term relative error is approximately 1.284×10^{-5} . The data also show

$$n(1 - 4a_n/p_n) = 1.739699, 1.940375, 2.110009, 2.286402, 2.386995, 2.463821$$

at the six listed indices, tending toward the proved limit $K/2 = 2.673017858628 \dots$

10.3 Testing the inverse at known sequence values

For $y = a_n$, the compatible inverse is exactly n . The leading Lambert approximation is s_0^2/B^2 ; the corrected one is the bracketed expression in (9.3).

n	Leading inverse minus n	Corrected inverse minus n
50	-0.598588435	-0.028604645
100	-0.560169154	-0.011441486
200	-0.530798365	-0.004479985
500	-0.502617905	-0.001257271
1000	-0.487447241	-0.000471103
2000	-0.476247422	-0.000174032

The nonvanishing limiting offset in the uncorrected Lambert approximation is explained by the constant term $2D_0/B^2$ in the inverse theorem. These values are numerical checks of already derived formulas.

11 Further research questions and concrete next steps

- 1. An all-orders coefficient expansion.** Use the general finite-cut identity (5.5) to extend the $O(n^{-7/4})$ remainder to arbitrary powers. The outstanding issue is not generating formal coefficients; it is uniform contour control of Q_m^{-1} and treatment of possible interior poles of rational approximants. A theorem organizing admissible cuts would turn the radial algorithm into an all-orders coefficient algorithm.
- 2. Explicit numerical error constants.** Make the product minor-arc constants and the cubic lower bound in the $m = 3$ proof effective. This would give certified numerical intervals for a_n and safe integer inversion, rather than asymptotic O statements with unspecified constants.

3. **A combinatorial explanation of the factor four.** Construct a direct relation between four copies of the L-convex class and the colored partitions counted by R . The proven inequality $p_n \geq 4a_n$ for $n \geq 1$ is suggestive, but an injection or decomposition is not supplied by the analytic proof. A description of the defect class could explain the limit $K/2$ combinatorially.
4. **The singularities of the boundary series.** Study $D(q) = L/R$ as an analytic object, not just a radial formal series. The recurrence, Wronskians, and positive tails give several exact representations. Determine its domain of continuation and the role of zeros of the finite denominators Q_m .
5. **Beyond-all-orders terms.** Separate exponentially small contributions to $\mathcal{A}(e^{-t})$ from the modular product and from the recurrence boundary corrections. Determine which roots of unity contribute to a coefficient transseries and whether the radial series has a useful summability or resurgence structure. None of these properties follows just from the computed rational coefficients.
6. **A reusable theorem for recurrence-defined area series.** Abstract the positive-transfer and finite-cut mechanism to other recurrences $y_{n+1} = a(q^n)y_n - b(q^n)y_{n-1}$ whose auxiliary generating functions are Euler products. State conditions under which a positive boundary defect is small enough for coefficient transfer. The present proof gives a worked example with all analytic and positivity obligations visible.
7. **Z-convex and higher-convexity area enumeration.** Investigate whether the numerically studied Z-convex area asymptotics [6] admit a related positive comparison. This is a proposed application, not an assertion that the L-convex recurrence or constant extends unchanged. Area and perimeter must remain distinct parameters.
8. **Formalization and OEIS integration.** Formalize the normalized transfer, coefficientwise order, Wronskian, and finite-cut identity first. These are algebraic and should be separated from the eta transformation and complex saddle analysis. An OEIS update should cite a reviewed version of the coefficient proof, rather than only a numerical output or the all-orders radial statement.

12 Conclusion and claim ledger

The principal result is a proof of a specific recorded conjecture, not a conjecture inferred from additional terms. The established generating function leads to a positive normalized recurrence; one auxiliary initial condition leads exactly to a colored-partition Euler product. Two boundary conditions produce the factor $1/4$. A positive coefficient squeeze proves the leading asymptotic and its first correction. A finite cut at index three, together with a full-circle error estimate, proves two further corrections and a relative $O(n^{-7/4})$ remainder.

The following boundaries summarize what can be reused without confusion.

Statement	Status in this manuscript
Area generating function and its enumerative interpretation	Cited input from prior work
Colored-partition product, positive squeeze, finite-cut identity	Proved here from the recurrence
Leading coefficient asymptotic and three correction terms	Proved with a complex remainder estimate
All-orders expansion for $\mathcal{A}(e^{-t}), t > 0$	Proved to every fixed radial order
Inverse expansion for a specified compatible interpolation	Proved; integer rounding treated separately
All-orders coefficient expansion and exponentially small sectors	Further research; not claimed
Finite checks through $n = 2000$ and symbolic arithmetic	Executed checks, not formal certification
Independent peer review or Lean verification	Not performed in this report

A Reproduction commands and files

The archive contains this standalone $\text{\LaTeX}$ source, the compiled PDF, two verification scripts, exact coefficient data, and their verification logs. Build the document with

```
pdflatex -interaction=nonstopmode -halt-on-error lconvex_asymptotics.tex
pdflatex -interaction=nonstopmode -halt-on-error lconvex_asymptotics.tex
```

Run the checks from the archive directory with

```
python verify.py --nmax 2000 --out checks
python verify_symbolic.py
```

The first command requires only the Python standard library. The second additionally requires SymPy and mpmath. The quadratic number of arithmetic operations in the coefficient checker is not a quadratic bit-complexity claim: the integer sizes grow with n . The code uses $O(N)$ large-integer storage for each of a fixed number of truncated series. No external data download is required for reproduction.

[write] Added 2 October 2026, batch 75. In the repository the source is `article.tex`, the scripts are in `code/`, and the contents of the delivered `checks/` directory are in `data/`; the compiled PDF of the archive is not shipped. `verify.py` writes into `-out` relative to the working directory, while `verify_symbolic.py` reads `coefficients.csv` from, and writes into, a `checks/` directory beside itself. The commands above therefore work only in the delivered layout; the report's README recreates it on a scratch copy. On 2 October 2026 that recipe reproduced `coefficients.csv` and both tables byte for byte and the other outputs apart from line endings.

B A short review checklist

A reader auditing the proof can concentrate on four potential failure points. First, retain the slope term when $z \rightarrow 1$ in (3.4); setting $(1-z)^2 C(z)$ to zero would give a false identity. Second, distinguish

coefficientwise positivity from positivity on the real interval; the proof of (4.3) establishes the stronger statements explicitly. Third, the finite-cut error is estimated on the *whole* coefficient circle using positive tails and the controlled full-circle integral of Q_3^{-2} . Fourth, the real-axis all-orders series is not used as a substitute for this complex estimate. These checks isolate the substantive logical steps of the argument.

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- [4] V. Kotěšovec, *Table of $n, a(n)$ for $n = 0, \dots, 2000$* , b-file for OEIS A126764. <https://oeis.org/A126764/b126764.txt>. Selected anchor values were checked against this source.
- [5] NIST Digital Library of Mathematical Functions, *Section 27.14: Unrestricted partitions*, especially 27.14.12–27.14.14 on Dedekind’s eta function and its transformation. <https://dlmf.nist.gov/27.14>. Accessed October 1, 2026.
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- [9] *Finite-prefix corrections and dual certificates for polyomino growth*, a report of the ProveIt research-report collection: `SetTheory/Cardinals/docs/reports/enumerative-combinatorics/polyomino-growth-finite-prefix-corrections`. Added in writing, batch 75.